

Given: Due:	Further Maths (Core Pure) HW 12	Name:
Chapter 7: Matrix Transformations (invariance)		
You MUST refer to class notes and complete personal study before attempting this homework. Attempt EVERY part of EVERY question. Check your work for errors or omissions before handing in.		
91% (A*)	76% (A)	66% (B)
56% (C)	50% (D)	40% (E)
Useful resources: Your notes from recent lessons & your text book <u>Although matrix arithmetic can be done on your calculator, show all steps to reach the answer & do use calculator for checking.</u>		
Questions - do these on lined paper		
1) The matrix $\begin{pmatrix} -0.28 & 0.96 \\ -0.96 & -0.28 \end{pmatrix}$ corresponds to a rotation R in the x-y plane. (i) State the centre of the rotation R, and find the angle of rotation (stating whether it is clockwise or anticlockwise). [3] The transformation S is the rotation R followed by reflection in the x-axis. (ii) Write down the matrix corresponding to reflection in the x-axis, and show that the matrix corresponding to the transformation S is $\begin{pmatrix} -0.28 & 0.96 \\ 0.96 & 0.28 \end{pmatrix}$. [3] (iii) Show that the invariant points of the transformation S lie on a straight line, and find the equation of this line. [4]		
2) The matrix $\begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix}$ represents a transformation, T. (i) Find the invariant points for the transformation T. [3] (ii) T is a transformation called a shear. The <i>line of shear</i> is the line of invariant points for the shear. The <i>factor</i> of a shear gives the distance a point is moved as a multiple of its perpendicular distance from the line of shear. What is the factor of the shear T? [4]		
3) The matrix $\begin{pmatrix} 3 & -2 \\ 4 & -6 \end{pmatrix}$ defines a transformation M of the (x, y) plane. (i) Show that the origin is the only invariant point of the transformation. [2] (ii) Find the two values of m for which $y = mx$ is an invariant line under the transformation. [4]		
END OF HOMEWORK Total = 23 marks		
Hand your work in with this sheet. Failure to meet your target grade will result in intervention support.		
File this question paper in your file, ready for revision and file inspection.		